Right-Triangle Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Trigonometric ratios**

A right triangle has legs of length 5 and 12 and hypotenuse 13. For the angle θ opposite the side of length 5, find:

a $\sin\theta = \frac{5}{13}$ **b** $\cos\theta = \frac{12}{13}$ **c** $\tan\theta = \frac{5}{12}$

(a) $= \frac{5}{13}$; (b) $= \frac{12}{13}$; (c) $= \frac{5}{12}$

2 **[Foundational]** • **Target: Trigonometric ratios**

Write the exact value of each expression:

a $\sin 30° = \frac{1}{2}$ **b** $\cos 45° = \frac{\sqrt{2}}{2}$ **c** $\tan 60° = \sqrt{3}$ **d** $\sin 60° = \frac{\sqrt{3}}{2}$

(a) $= \frac{1}{2}$; (b) $= \frac{\sqrt{2}}{2}$; (c) $= \sqrt{3}$; (d) $= \frac{\sqrt{3}}{2}$

3 **[Foundational]** • **Target: Trigonometric ratios**

A right triangle has legs of length 7 and 24.

a Find the length of the hypotenuse. $\sqrt{7^2 + 24^2} = \sqrt{625} = 25$.

b Find the measure of the angle opposite the side of length 7, to the nearest hundredth of a degree. $\theta = \tan^{-1}\left(\frac{7}{24}\right) \approx 16.26°$.

(a) $\sqrt{7^2 + 24^2} = \sqrt{625} = 25$.; (b) $\theta = \tan^{-1}\left(\frac{7}{24}\right) \approx 16.26°$.

4 **[Foundational]** • **Target: Applications**

A ladder leans against a wall, making an angle of $62°$ with the ground. The foot of the ladder is 4.5 m from the base of the wall. How high up the wall does the ladder reach? Round to two decimal places.

$h = 4.5\tan 62° \approx 4.5 \times 1.8807 \approx 8.46$ m.

5 **[Foundational]** • **Target: Applications**

From the top of a cliff 50 m above sea level, the angle of depression to a boat is $18°$. How far is the boat from the base of the cliff? Round to one decimal place.

$d = \frac{50}{\tan 18°} \approx \frac{50}{0.3249} \approx 153.9$ m.

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6 [Foundational] • Target: Applications

A kite string is 85 m long and makes an angle of $54°$ with the horizontal ground. Assuming the string is straight, find the height of the kite above the ground, to the nearest metre.

$h = 85\sin 54° \approx 85 \times 0.8090 \approx 69$ m.

7 [Foundational] • Target: Applications

Two buildings are 40 m apart. From the top of the shorter building (height 25 m), the angle of elevation to the top of the taller building is $32°$. Find the height of the taller building, to one decimal place.

Rise $= 40\tan 32° \approx 40(0.6249) \approx 25.0$ m, so the taller building is about $25 + 25.0 = 50.0$ m.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Applications

A surveyor stands 120 m from the base of a tower and measures the angle of elevation to the top as $37°$. Find the height of the tower, to the nearest tenth of a metre.

$h = 120\tan 37° \approx 120(0.7536) \approx 90.4$ m.

9 [Intermediate] • Target: Solving right triangles

In right triangle ABC with the right angle at C, $A = 35°$ and the hypotenuse $AB = 20$. Find:

a side BC (opposite A) $BC = 20\sin 35° \approx 11.47$.

b side AC (adjacent to A) $AC = 20\cos 35° \approx 16.38$.

c angle B $B = 90° - 35° = 55°$.

(a) $BC = 20\sin 35° \approx 11.47$.; (b) $AC = 20\cos 35° \approx 16.38$.; (c) $B = 90° - 35° = 55°$.

10 [Intermediate] • Target: Solving right triangles

In right triangle PQR with the right angle at R, the two legs are $PR = 9$ and $QR = 12$. Find angle P (opposite QR) to the nearest tenth of a degree, and the length of the hypotenuse.

$\tan P = \frac{12}{9}$, so $P = \tan^{-1}\left(\frac{12}{9}\right) \approx 53.1°$. Hypotenuse $= \sqrt{9^2 + 12^2} = 15$.

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11 [Intermediate] • Target: Angles of elevation and depression

A drone hovers at a height of 60 m. The angle of depression from the drone to a landmark on the ground is $27°$. Find the horizontal distance from the point directly below the drone to the landmark, to the nearest metre.

$d = \frac{60}{\tan 27°} \approx \frac{60}{0.5095} \approx 118$ m.

12 [Intermediate] • Target: Angles of elevation and depression

From a lighthouse 45 m tall, the angle of depression to a ship is $12°$, and a second ship is spotted further out with angle of depression $7°$. Find the distance between the two ships, to the nearest metre.

Distances from the base: $d_1 = \frac{45}{\tan 12°} \approx 211.7$ m; $d_2 = \frac{45}{\tan 7°} \approx 366.5$ m. Distance between ships $\approx 366.5 - 211.7 \approx 154.8$ m.

13 [Intermediate] • Target: The reciprocal trig ratios

For a right triangle with opposite $= 3$, adjacent $= 4$, hypotenuse $= 5$ (relative to angle θ), find:

a $\csc\theta = \frac{5}{3}$ **b** $\sec\theta = \frac{5}{4}$ **c** $\cot\theta = \frac{4}{3}$

(a) $= \frac{5}{3}$; (b) $= \frac{5}{4}$; (c) $= \frac{4}{3}$

14 [Intermediate] • Target: The reciprocal trig ratios

Explain why $\csc\theta = \frac{1}{\sin\theta}$, and use this to find $\csc(30°)$.

$\csc\theta$ is defined as the reciprocal of $\sin\theta$ (hypotenuse over opposite, instead of opposite over hypotenuse). $\csc(30°) = \frac{1}{\sin(30°)} = \frac{1}{1/2} = 2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Multi-step surveying problems • Cross-Lesson: 6.02 Polar Equations and Graphs

A surveyor at point A measures the angle of elevation to a mountain peak as $22°$. Walking 500 m directly toward the mountain to point B, the angle of elevation increases to $35°$. Find the height of the mountain, to the nearest metre. (Hint: let h be the height and x the distance from B to the base; set up two equations.)

From B: $h = x\tan 35°$. From A: $h = (x + 500)\tan 22°$. Setting equal: $x\tan 35° = (x + 500)\tan 22°$. $x(0.7002) = (x + 500)(0.4040)$. $0.7002x - 0.4040x = 202$. $0.2962x = 202 \Rightarrow x \approx 682.0$. $h = 682.0(0.7002) \approx 477.5$ m.

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16 **[Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.01 Exponential Functions and Their Graphs**

In a 30-60-90 triangle, the shortest side (opposite the $30°$ angle) has length 5. Find the lengths of the other two sides in exact form.

Side ratios are $1 : \sqrt{3} : 2$. Side opposite $60°$: $5\sqrt{3}$. Hypotenuse: 10.

17 **[Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.02 Logarithms and Their Properties**

In a 45-45-90 triangle, each leg has length 7. Find the hypotenuse in exact form.

Hypotenuse $= 7\sqrt{2}$ (using the ratio $1 : 1 : \sqrt{2}$).

18 **[Challenging] • Target: Trigonometric ratios beyond the first quadrant (preview) • Cross-Lesson: 7.03 Exponential and Logarithmic Equations**

A right triangle used to define $\sin\theta$ and $\cos\theta$ only works directly for acute angles θ. Explain why a different (coordinate-based) definition is needed to extend these ratios to obtuse or negative angles.

A right triangle can only have angles strictly between $0°$ and $90°$ (since it already has one $90°$ angle and angles sum to $180°$), so triangle ratios cannot directly define sine or cosine for angles at or beyond $90°$. Extending to all angles requires the unit-circle definition, where $\cos\theta$ and $\sin\theta$ are the x- and y -coordinates of a point on the unit circle at angle θ.

19 **[Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.04 Applications: Growth and Decay**

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

20 **[Challenging] • Target: Trigonometric ratios beyond the first quadrant (preview) • Cross-Lesson: 8.01 Rational Functions and Asymptotes**

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse AB.

We can use the Pythagorean theorem to find the length of the hypotenuse in a right-angled triangle. The theorem states that $a^2 + b^2 = c^2$, where a and b are the lengths of the two shorter sides (legs) and c is the length of the hypotenuse. In this triangle, $AC = 10$ and $BC = 24$. Therefore, we have: $AB^2 = AC^2 + BC^2$ $AB^2 = 10^2 + 24^2$ $AB^2 = 100 + 576$ $AB^2 = 676$ $AB = \sqrt{676}$ $AB = 26$ The length of the hypotenuse AB is 26.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Six Trigonometric Ratios

In a right triangle with acute angle θ, the side opposite θ is 5 and the hypotenuse is 13. Find $\cos\theta$.

a. 12/13 ✓

b. 5/12

c. 13/12

d. 12/5

Adjacent side $a = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = 12$. $\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = 12/13$.

22 [Foundational] • Target: The Six Trigonometric Ratios

For the same triangle ($extopp = 5$, adj $= 12$, hyp $= 13$), find $\sec\theta$.

a. 13/12 ✓

b. 13/5

c. 12/13

d. 12/5

$\sec\theta = \frac{1}{\cos\theta} = \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{13}{12}$.

23 [Foundational] • Target: The Six Trigonometric Ratios

If $\tan\theta = 3/4$ in a right triangle, find $\csc\theta$.

a. 5/3 ✓

b. 5/4

c. 4/3

d. 3/5

Opposite $= 3$, adjacent $= 4 \Rightarrow$ hypotenuse $= 5$. $\csc\theta = \frac{\text{hypotenuse}}{\text{opposite}} = 5/3$.

24 [Foundational] • Target: Special Right Triangles (30-60-90 and 45-45-90)

In a $30^\circ - 60^\circ - 90^\circ$ triangle, if the shorter leg is 4, find the lengths of the longer leg and the hypotenuse.

a. **Longer leg** $= 4\sqrt{3}$**, Hypotenuse** $= 8$ ✓

b. Longer leg $= 8$, Hypotenuse $= 4\sqrt{3}$

c. Longer leg $= 4\sqrt{2}$, Hypotenuse $= 8$

d. Longer leg $= 6$, Hypotenuse $= 8$

Ratio of sides opposite $30^\circ : 60^\circ : 90^\circ$ is $1 : \sqrt{3} : 2$. With short leg 4: longer leg is $4\sqrt{3}$ and hypotenuse is 8.

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25 [Foundational] • Target: Special Right Triangles (30-60-90 and 45-45-90)

In a $45^\circ - 45^\circ - 90^\circ$ triangle, if the hypotenuse is 10, find the length of each leg.

a. $5\sqrt{2}$ ✓

b. 5

c. $10\sqrt{2}$

d. $5\sqrt{3}$

Hypotenuse $= s\sqrt{2} = 10 \Rightarrow s = \frac{10}{\sqrt{2}} = 5\sqrt{2}$.

26 [Foundational] • Target: Special Right Triangles (30-60-90 and 45-45-90)

What is the exact value of $\sec(60^\circ) + \tan(45^\circ)$?

a. 3 ✓

b. $2 + \sqrt{3}$

c. $1 + \sqrt{3}$

d. 2

$\cos(60^\circ) = 1/2 \Rightarrow \sec(60^\circ) = 2$. $\tan(45^\circ) = 1$. Sum $= 2 + 1 = 3$.

27 [Foundational] • Target: Cofunction Identities and Reciprocals

Which cofunction identity correctly relates sine and cosine for complementary acute angles?

a. $\sin(90^\circ - \theta) = \cos\theta$ ✓

b. $\sin(90^\circ - \theta) = -\cos\theta$

c. $\sin(90^\circ - \theta) = \sin\theta$

d. $\sin(90^\circ - \theta) = \tan\theta$

In a right triangle, the two acute angles are complementary, so the side opposite one angle is adjacent to the other: $\sin(90^\circ - \theta) = \cos\theta$.

28 [Foundational] • Target: Cofunction Identities and Reciprocals

If $\cos(28^\circ) \approx 0.8829$, what is $\sin(62^\circ)$?

a. 0.8829 ✓

b. 0.1171

c. 0.4695

d. 1.1326

Since $28^\circ + 62^\circ = 90^\circ$, $\sin(62^\circ) = \cos(90^\circ - 62^\circ) = \cos(28^\circ) \approx 0.8829$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cofunction Identities and Reciprocals

Simplify $\cot(90^\circ - \theta) \cdot \cot\theta$.

a. 1 ✓

b. $\tan^2\theta$

c. $\cot^2\theta$

d. 0

By cofunction identity, $\cot(90^\circ - \theta) = \tan\theta$. Then $\tan\theta \cdot \cot\theta = \tan\theta \cdot \frac{1}{\tan\theta} = 1$.

30 [Intermediate] • Target: Angles of Elevation and Depression

From a point 50 ft from the base of a building, the angle of elevation to the top is 60°. How tall is the building?

a. $50\sqrt{3}$ ft ✓

b. 100 ft

c. 50 ft

d. $25\sqrt{3}$ ft

$\tan(60^\circ) = \frac{h}{50} \Rightarrow h = 50\tan(60^\circ) = 50\sqrt{3}$ ft.

31 [Intermediate] • Target: Angles of Elevation and Depression

A lighthouse keeper 120 ft above sea level spots a boat at an angle of depression of 30°. How far is the boat from the base of the lighthouse?

a. $120\sqrt{3}$ ft ✓

b. $60\sqrt{3}$ ft

c. 240 ft

d. 120 ft

The angle of elevation from the boat equals the angle of depression: 30°.
$\tan(30^\circ) = \frac{120}{d} \Rightarrow d = \frac{120}{\tan(30^\circ)} = \frac{120}{1/\sqrt{3}} = 120\sqrt{3}$ ft.

32 [Intermediate] • Target: Angles of Elevation and Depression

A $20-$ft ladder leans against a wall reaching a height of 16 ft. What is $\sin\theta$ where θ is the angle the ladder makes with the ground?

a. $4/5$ ✓

b. $3/5$

c. $4/3$

d. $3/4$

$\sin\theta = \frac{\text{height}}{\text{length}} = \frac{16}{20} = \frac{4}{5}$.

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33 [Intermediate] • Target: Trigonometric ratios

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$ ✓

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

We are given $\cos\alpha = \frac{3}{5}$ and α is in the first quadrant. Using the identity $\sin^2\alpha + \cos^2\alpha = 1$, we find $\sin\alpha = \sqrt{1-(\frac{3}{5})^2} = \sqrt{1-\frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$ (since α is in the first quadrant, $\sin\alpha$ is positive). We are given $\sin\beta = -\frac{4}{5}$ and β is in the third quadrant. Using the identity $\sin^2\beta + \cos^2\beta = 1$, we find.

34 [Intermediate] • Target: Applications

If $\cos\theta = -\frac{1519}{1681}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1600}{1681}$

b. $-\frac{40}{41}$

c. $\frac{40}{41}$ ✓

d. $\frac{441}{410}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-(-\frac{1519}{1681})}{2}} = \frac{40}{41}$.

35 [Intermediate] • Target: Solving right triangles

If $\cos\theta = -\frac{3479}{3721}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{661}{610}$

b. $-\frac{60}{61}$

c. $\frac{60}{61}$ ✓

d. $\frac{3600}{3721}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-(-\frac{3479}{3721})}{2}} = \frac{60}{61}$.

36 [Intermediate] • Target: Angles of elevation and depression

If $\cos\theta = \frac{3479}{3721}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $-\frac{11}{61}$

b. $\frac{171}{610}$

c. $\frac{121}{3721}$

d. $\frac{11}{61}$ ✓

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-(\frac{3479}{3721})}{2}} = \frac{11}{61}$.

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37 [Intermediate] • Target: The reciprocal trig ratios

If $\cos\theta = -\frac{119}{169}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{133}{130}$

b. $-\frac{12}{13}$

c. $\frac{12}{13}$ ✓

d. $\frac{144}{169}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(-\frac{119}{169}\right)}{2}} = \frac{12}{13}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multi-step surveying problems • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

If $\cos\theta = -\frac{527}{625}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{576}{625}$

b. $\frac{24}{25}$ ✓

c. $-\frac{24}{25}$

d. $\frac{53}{50}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(-\frac{527}{625}\right)}{2}} = \frac{24}{25}$.

39 [Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.04 Applications: Growth and Decay

If $\cos\theta = -\frac{161}{289}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{225}{289}$

b. $-\frac{15}{17}$

c. $\frac{167}{170}$

d. $\frac{15}{17}$ ✓

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(-\frac{161}{289}\right)}{2}} = \frac{15}{17}$.

40 [Challenging] • Target: Trigonometric ratios beyond the first quadrant (preview) • Cross-Lesson: 8.01 Rational Functions and Asymptotes

If $\cos\theta = \frac{1519}{1681}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{9}{41}$ ✓

b. $-\frac{9}{41}$

c. $\frac{81}{1681}$

d. $\frac{131}{410}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{1519}{1681}\right)}{2}} = \frac{9}{41}$.

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41 [Challenging] • Target: The Six Trigonometric Ratios • Cross-Lesson: 8.02 Graphing Rational Functions

If $\cos\theta = -\frac{41}{841}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{239}{290}$

b. $\frac{441}{841}$

c. $\frac{21}{29}$ ✓

d. $-\frac{21}{29}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(-\frac{41}{841}\right)}{2}} = \frac{21}{29}$.

42 [Challenging] • Target: Special Right Triangles (30-60-90 and 45-45-90) • Cross-Lesson: 8.03 Introduction to Limits

If $\cos\theta = \frac{119}{169}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{25}{169}$

b. $-\frac{5}{13}$

c. $\frac{5}{13}$ ✓

d. $\frac{63}{130}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{119}{169}\right)}{2}} = \frac{5}{13}$.

43 [Challenging] • Target: Cofunction Identities and Reciprocals • Cross-Lesson: 9.01 Arithmetic Sequences and Series

If $\cos\theta = \frac{527}{625}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{7}{25}$ ✓

b. $\frac{19}{50}$

c. $-\frac{7}{25}$

d. $\frac{49}{625}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{527}{625}\right)}{2}} = \frac{7}{25}$.

44 [Challenging] • Target: Angles of Elevation and Depression • Cross-Lesson: 9.02 Geometric Sequences and Series

If $\cos\theta = \frac{161}{289}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{8}{17}$ ✓

b. $\frac{64}{289}$

c. $\frac{97}{170}$

d. $-\frac{8}{17}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{161}{289}\right)}{2}} = \frac{8}{17}$.

45 [Challenging] • Target: Trigonometric ratios • Cross-Lesson: 9.03 The Binomial Theorem

If $\cos\theta = \frac{41}{841}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{229}{290}$

b. $\frac{20}{29}$ ✓

c. $-\frac{20}{29}$

d. $\frac{400}{841}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{41}{841}\right)}{2}} = \frac{20}{29}$.

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